

# SecurePulse Mobile — Master Project & Documentation Audit

| Audit Specification        | Audit Value / Status                                            |
|----------------------------|-----------------------------------------------------------------|
| **Audit Date**             | **2026-09-01**                                                  |
| **Repository Name**        | `NATTOMR/securepulse-mobile`                                    |
| **Application ID**         | `com.securepulse.mobile` (version `1.0.0+1`)                    |
| **Repository Scope**       | Flutter Mobile Client Application & Cloud API Integration Layer |
| **Automated Test Results** | <div></div> **33 / 33 Tests Passed (100% Success Rate)**        |
| **Git Working Tree**       | <div></div> **Clean (`main` branch)**                           |

## 1. Project Overview

**SecurePulse Mobile** is a specialized mobile cybersecurity operations and incident triage platform built with Flutter 3.x and Dart 3.x. The platform serves Security Operations Center (SOC) analysts, incident responders, and security administrators by providing real-time threat telemetry, automated SAST code vulnerability auditing, conversational AI remediation assistance, and cryptographic PDF compliance report generation.

The mobile client is engineered with a **Dual Operation Architecture**:

- **Demo / Standalone Mode:** Fully functional offline simulation environment with deterministic mock security data, simulated Wazuh SIEM alerts, CVE remediation playbooks, and precomputed posture scores for zero-latency demonstrations and offline use.
- **Live Backend Mode:** Production network client connecting via REST and WebSockets to the external FastAPI backend (deployed on Render Cloud at `https://secureguard-backend-7eqm.onrender.com` or local developer endpoints).

## 2. Current Architecture

```

graph TD
    subgraph Mobile_Presentation_Layer [Flutter 3.x / Dart 3.x]
        Shell[GoRouter 5-Tab Shell Navigation]
        Theme[AppTheme: Cyber Dark Obsidian & Clean Light]
        Screens[Dashboard, Repositories, AI Assistant, Alerts, Settings, Reports, Scans, Profile]
    end

    subgraph State_Business_Logic_Layer
        Riverpod[Riverpod 2.5.1 State Providers & Stream Controllers]
        Domain[Domain Models: User, Dashboard, Alert, Repository, Scan, Finding, AI Message]
    end

    subgraph Core_Infrastructure_Layer
        Bio[BiometricService: Hardware Face ID / Fingerprint]
        PDF[ReportPdfService: Vector PDF Engine + SHA256 Audit Stamp]
        Storage[SecureStorageService: AES-256 Keychain + HiveStorageService: Local Cache]
        NetClient[ApiClient: Dio 5.4.1 with JWT Interceptors & Timeout Gates]
        WS[WebSocketService: Real-Time Stream + Reconnect + Polling Fallback]
    end

    subgraph Operation_Mode_Dispatcher
        ModeCheck{AppConfig.isDemoMode}
        MockLayer[Mock Data Repositories: Auth, Dashboard, Alerts, Repos, Scans, AI]
        LiveGateway[FastAPI Cloud Gateway on Render / Localhost]
    end

    subgraph External_Backend_Services [External Cloud Infrastructure]
        FastAPIEngine[FastAPI Python Backend]
        WazuhEngine[Wazuh SIEM Connector]
        SemgrepEngine[Semgrep SAST Engine]
        FirebaseEngine[Firebase Cloud Messaging FCM]
    end

    Shell --> Riverpod
    Screens --> Riverpod
    Riverpod --> Domain
    Riverpod --> Bio & PDF & Storage
    Riverpod --> NetClient & WS
    NetClient & WS --> ModeCheck
    ModeCheck -- True (Demo) --> MockLayer
    ModeCheck -- False (Live) --> LiveGateway
    LiveGateway --> FastAPIEngine
    FastAPIEngine --> WazuhEngine & SemgrepEngine & FirebaseEngine

```

### 3. Technology Stack

#### 3.1. Client-Side Framework & Runtime

- **Framework:** Flutter SDK `flutter` (Dart `>=3.0.0 <4.0.0`)
- **State Management:** `flutter_riverpod` `^2.5.1`
- **Declarative Routing:** `go_router` `^13.2.0`
- **HTTP Networking:** `dio` `^5.4.1`
- **Real-Time Streaming:** `web_socket_channel` `^3.0.1`
- **Secure Storage:** `flutter_secure_storage` `^9.0.0` (Android Keystore / iOS Keychain)
- **Local Offline Caching:** `hive` `^2.2.3`, `hive_flutter` `^1.1.0`, `shared_preferences` `^2.2.2`
- **Biometric Hardware Integration:** `local_auth` `^2.1.8`
- **Push Notifications:** `firebase_core` `^2.27.0`, `firebase_messaging` `^14.7.19`
- **Document Engine:** `pdf` `^3.10.8`, `path_provider` `^2.1.2`, `share_plus` `^7.2.2`
- **UI & Data Visualization:** `fl_chart` `^0.66.2`, `flutter_animate` `^4.5.0`, `google_fonts` `^6.1.0`, `shimmer` `^3.0.0`
- **Code Quality & Analysis:** `flutter_lints` `^3.0.0`, `flutter_test`

#### 3.2. Containerization & Deployment

- **Web Container:** Single-stage `nginx:alpine` Docker image serving a pre-built Flutter web bundle ( `build/web` ) on port 80. The `flutter build web --release` step must be executed separately before the Docker image build.
  - **Android Release Output:** Compiled standalone release binary `securepulse-release.apk` (63.6 MB).
  - **Cloud API Target:** Render Web Service ( <https://secureguard-backend-7eqm.onrender.com> ).
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#### 4. Implemented Features

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| Capability                                    | Status                                                                                             | Source Location                                                                                                                                                                                                                                                                                                                                                                                                                                             | Verification & Implementation Evidence                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| **Dual Operation Mode (Demo / Live)**         | <br>IMPLEMENTED   | [lib/core/config/app_config.dart:28](file:///e:/SOC%20projects/securepulse-mobile/lib/core/config/app_config.dart#L28),<br>[lib/features/settings/presentation/settings_screen.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/settings/presentation/settings_screen.dart)                                                                                                                                                              | Dynamic toggle backed by 'SharedPreferences' ('sg_is_demo_mode'). Repositories dynamically route requests between local mock datasets and live HTTP/WS network clients.                                                                                                                                                                                                                                                 |
| **Riverpod State Management**                 | <br>IMPLEMENTED   | [lib/providers/app_providers.dart](file:///e:/SOC%20projects/securepulse-mobile/lib/providers/app_providers.dart)                                                                                                                                                                                                                                                                                                                                           | Clean provider tree providing auth state, posture scores, repository lists, scan histories, SIEM alerts, and real-time WebSocket streams.                                                                                                                                                                                                                                                                               |
| **5-Tab Declarative Shell Routing**           | <br>IMPLEMENTED   | [lib/core/router/app_router.dart:59-123]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/router/app_router.dart#L59-L123)                                                                                                                                                                                                                                                                                                                         | 'GoRouter' shell route hosting 5 primary persistent bottom tabs ('Dashboard', 'Repositories', 'AI Assistant', 'SOC Alerts', 'Settings') plus modal subroutes for details.                                                                                                                                                                                                                                               |
| **Executive Posture Dashboard**               | <br>IMPLEMENTED   | [lib/features/dashboard/presentation/dashboard_screen.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/dashboard/presentation/dashboard_screen.dart)                                                                                                                                                                                                                                                                                     | Animated posture score gauge (88–94%), vulnerability breakdown donut chart ('fl_chart'), live service status badges, and quick-action docks.                                                                                                                                                                                                                                                                            |
| **Hardware Biometric Authentication**         | <br>IMPLEMENTED   | [lib/core/services/biometric_service.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/services/biometric_service.dart),<br>[android/app/src/main/AndroidManifest.xml:5-6]<br>(file:///e:/SOC%20projects/securepulse-mobile/android/app/src/main/AndroidManifest.xml#L5-L6)                                                                                                                                                                   | 'local_auth' hardware biometric challenge (Face ID & Fingerprint) for session locking and authentication pre-checks.                                                                                                                                                                                                                                                                                                    |
| **Encrypted Local Storage**                   | <br>IMPLEMENTED   | [lib/core/storage/secure_storage_service.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/storage/secure_storage_service.dart),<br>[lib/core/storage/hive_storage_service.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/storage/hive_storage_service.dart)                                                                                                                                                                 | 'FlutterSecureStorage' (AES-256 / RSA device keychain) storing JWT tokens and custom backend URLs; 'HiveStorageService' caching offline app state in 'securepulse_cache' box.                                                                                                                                                                                                                                           |
| **REST API Client with Interceptors**         | <br>IMPLEMENTED | [lib/core/network/api_client.dart](file:///e:/SOC%20projects/securepulse-mobile/lib/core/network/api_client.dart), [lib/core/error/api_exception.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/error/api_exception.dart)                                                                                                                                                                                                                  | Dio 5.4.1 client with automatic 'Bearer <token>' header injection, 12-second timeout limits, dynamic base URL mutation, and classified ['ApiException']<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/error/api_exception.dart) error mappings.                                                                                                                                                             |
| **Real-Time WebSocket Incident Stream**       | <br>IMPLEMENTED | [lib/core/network/websocket_service.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/network/websocket_service.dart)                                                                                                                                                                                                                                                                                                                         | Connects to '/ws/alerts', automatically converts 'http:// -> ws://' and 'https:// -> wss://', maintains stream broadcasting, and falls back to HTTP polling if blocked.                                                                                                                                                                                                                                                 |
| **Codebase & SAST Vulnerability Audits**      | <br>IMPLEMENTED | [lib/features/repositories/data/repository_repository.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/repositories/data/repository_repository.dart),<br>[lib/features/scans/scans_screen.dart](file:///e:/SOC%20projects/securepulse-mobile/lib/features/scans/scans_screen.dart)                                                                                                                                                       | Repository list with language tags, branch info, health grades (A–F), finding counts, and on-demand scan triggers ('/v1/repositories/{id}/scan').                                                                                                                                                                                                                                                                       |
| **AI Cybersecurity Assistant**                | <br>IMPLEMENTED | [lib/features/ai/data/ai_repository.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/ai/data/ai_repository.dart),<br>[lib/features/ai/presentation/ai_assistant_screen.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/ai/presentation/ai_assistant_screen.dart)                                                                                                                                                     | Conversational security assistant. Generates local playbooks for CVE-2024-3094, SQLi, and secret exposure in Demo mode, or forwards to '/v1/ai/chat' in Live mode.                                                                                                                                                                                                                                                      |
| **SOC SIEM Alert Triage & Wazuh Manager API** | <br>IMPLEMENTED | [lib/features/alerts/data/alerts_repository.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/alerts/data/alerts_repository.dart),<br>[lib/features/alerts/data/wazuh_repository.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/alerts/data/wazuh_repository.dart),<br>[lib/features/alerts/domain/wazuh_models.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/alerts/domain/wazuh_models.dart) | Incident stream filtered by severity (Critical, High, Medium, Low), detailed triage with origin IP, destination port, raw syslog text, and status update actions ('/v1/soc/alerts/{id}/status'). Includes full bidirectional Wazuh Manager API integration for agent inventory inspection ('WazuhAgentModel'), daemon health probes ('wazuh-analysisd', 'wazuh-remoted', 'wazuh-modulesd'), and remote restart actions. |
| **GitHub OAuth2 Deep-Linking Integration**    | <br>IMPLEMENTED | [lib/core/services/github_oauth_service.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/services/github_oauth_service.dart),<br>[lib/features/auth/data/auth_repository.dart]<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/auth/data/auth_repository.dart),<br>[android/app/src/main/AndroidManifest.xml:40-48]<br>(file:///e:/SOC%20projects/securepulse-mobile/android/app/src/main/AndroidManifest.xml#L40-L48)         | Complete PKCE-compatible OAuth2 consent loop. Generates high-entropy cryptographic CSRF state tokens, opens system browser via 'url_launcher', captures 'securepulse://oauth/callback' via 'app_links', and exchanges authorization code via 'POST /v1/auth/github'.                                                                                                                                                    |

| Capability                                 | Status                                                                                          | Source Location                                                                                                                                                                                                                                                                                                                                           | Verification & Implementation Evidence                                                                                                                                                                                                                                                                                             |
|--------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| **Offline Action & Mutation Queue Engine** |  IMPLEMENTED   | <code>[lib/core/services/offline_queue_service.dart]</code><br><code>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/services/offline_queue_service.dart),</code><br><code>[lib/features/alerts/data/alerts_repository.dart]</code><br><code>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/alerts/data/alerts_repository.dart)</code> | Durably captures analyst mitigation actions (e.g. alert status updates, remediation triggers) in local encrypted Hive cache when offline and automatically flushes them with exponential retry handling upon network reconnect ('connectivity_plus').                                                                              |
| **Vector PDF Compliance Generator**        |  IMPLEMENTED   | <code>[lib/core/services/report_pdf_service.dart]</code><br><code>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/services/report_pdf_service.dart),</code><br><code>[lib/features/reports/reports_screen.dart]</code><br><code>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/reports/reports_screen.dart)</code>                     | Pure vector PDF engine compiling official compliance audit reports for SOC 2 Type II, ISO 27001, PCI-DSS v4.0, and HIPAA. Reports embed a deterministic, cryptographically computed SHA-256 audit digest ('crypto' package) and time-stamped audit ID for complete non-repudiation and evidence integrity.                         |
| **Environment Switcher & Diagnostics**     |  IMPLEMENTED   | <code>[lib/features/settings/presentation/settings_screen.dart]</code><br><code>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/settings/presentation/settings_screen.dart)</code>                                                                                                                                                             | Multi-environment switcher (Render Cloud, Android Emulator '10.0.2.2', Localhost '127.0.0.1', Custom URL) with live HTTP health ping diagnostics.                                                                                                                                                                                  |
| **Dual Theme System (Dark / Light)**       |  IMPLEMENTED   | <code>[lib/core/theme/app_theme.dart]</code> <code>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/theme/app_theme.dart)</code>                                                                                                                                                                                                                    | Material 3 Cyber Dark Obsidian ('#0A0E1A') and Clean Light ('#F8F8F8') themes with Google Fonts Inter and custom semantic color tokens.                                                                                                                                                                                            |
| **Automated Test Suite**                   |  IMPLEMENTED   | <code>[test/api_integration_test.dart]</code> <code>(file:///e:/SOC%20projects/securepulse-mobile/test/api_integration_test.dart),</code> <code>[test/widget_test.dart]</code><br><code>(file:///e:/SOC%20projects/securepulse-mobile/test/widget_test.dart)</code>                                                                                       | 33/33 automated tests verifying network contracts, mode isolation, repository fallback behavior, WebSocket URL conversion, GitHub OAuth2 URL & CSRF parsing, SHA-256 PDF audit digest calculation & avalanche verification, Wazuh agent/daemon models & remote actions, Offline Queue durability & auto-flush, and widget pumping. |
| **Web Deployment Dockerfile**              |  IMPLEMENTED | <code>[Dockerfile]</code> <code>(file:///e:/SOC%20projects/securepulse-mobile/Dockerfile)</code>                                                                                                                                                                                                                                                          | Single-stage Alpine Nginx container ('nginx:alpine') serving a pre-built Flutter web bundle ('build/web') on port 80. The 'flutter build web --release' command must be executed prior to Docker image build. The Dockerfile does not contain a Flutter build stage.                                                               |

## 5. Partially Implemented Features

| Capability                         | Status                                                                                                    | Source Location                                                                                                                                                                                   | Actual State & Current Limitations                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| **Firebase Cloud Messaging (FCM)** |  PARTIALLY IMPLEMENTED | <code>[lib/core/services/notification_service.dart]</code><br><code>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/services/notification_service.dart)</code>                             | Background and foreground push handlers, topic subscriptions ('soc_critical', 'wazuh_alerts', 'semgrep_findings'), and permission requests are fully coded. A 'google-services.json' file (678 bytes) is present at 'android/app/google-services.json' as a placeholder configuration. Full remote push delivery requires a valid Firebase project configuration to replace this placeholder. The service gracefully falls back to local in-app stream broadcasting without crashing. |
| **Direct SOAR Edge Execution**     |  PARTIALLY IMPLEMENTED | <code>[lib/features/alerts/presentation/alert_detail_screen.dart]</code><br><code>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/alerts/presentation/alert_detail_screen.dart)</code> | "Quarantine IP" and "Acknowledge" buttons update alert status via '/v1/soc/alerts/{id}/status'. Direct cloud firewall / router rule execution is delegated to backend event listeners rather than triggered via direct mobile-to-firewall API.                                                                                                                                                                                                                                        |

## 6. Planned Features

| Capability                                | Status    | Target Roadmap Phase | Description                                                                                                                   |
|-------------------------------------------|-----------|----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| **Two-Way SIEM / SOAR Connectors**        | ● PLANNED | Phase 4              | Bidirectional webhook synchronization for Splunk, Elastic SIEM, Microsoft Sentinel, and Jira Security ticketing.              |
| **In-App Custom Semgrep Rule Editor**     | ● PLANNED | Phase 4              | In-app YAML authoring interface with syntax validation for custom Semgrep and Sigma detection rules.                          |
| **Multi-Tenant / Organization Switching** | ● PLANNED | Phase 3              | Ability to seamlessly switch between multiple enterprise security clusters and client tenants within a single mobile session. |
| **WearOS & Apple Watch Companion**        | ● PLANNED | Phase 5              | Wearable companion application for critical severity paging, alert acknowledgment, and biometric quick-triage.                |

## 7. Not Implemented Features ❌

| Capability                              | Status            | Rationale & Code Inspection Finding                                                                                                                                            |
|-----------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| **Zero-Trust Client mTLS Certificates** | ❌ NOT IMPLEMENTED | Mutual TLS certificate provisioning per individual mobile client hardware certificate is not implemented in 'ApiClient'. Communication relies on TLS 1.3 with JWT Bearer auth. |
| **Voice-Activated SOC Commands**        | ❌ NOT IMPLEMENTED | Speech-to-command natural language processing is not implemented in the application.                                                                                           |

## 8. Verified API Endpoints

The mobile client is coded and tested against the following REST and WebSocket endpoint contracts defined in [ `ApiEndpoints` ] (file:///e:/SOC%20projects/securepulse-mobile/lib/core/network/api\_endpoints.dart):

| Endpoint                    | Method    | Authentication         | Request Payload                       | Response Model / Payload                      | Client Handling Class                                                                                                                                            |
|-----------------------------|-----------|------------------------|---------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| /health`                    | GET`      | None                   | None                                  | { "status": "ok", "version": "1.0.0" }        | Diagnostic health ping in `SettingsScreen`                                                                                                                       |
| /v1/auth/login`             | POST`     | None                   | { "email": "...", "password": "..." } | { "token": "...", "user": {...} }             | [ `AuthRepositoryImpl.login()`<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/auth/data/auth_repository.dart)                                     |
| /v1/auth/me`                | GET`      | Bearer <JWT>`          | None                                  | UserModel` JSON                               | [ `AuthRepositoryImpl.loginWithBiometrics()`<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/auth/data/auth_repository.dart)                       |
| /v1/dashboard/summary`      | GET`      | Bearer <JWT>`          | None                                  | DashboardModel` JSON (score, counts, systems) | [ `DashboardRepositoryImpl.getDashboardSummary()`<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/dashboard/data/dashboard_repository.dart)        |
| /v1/repositories`           | GET`      | Bearer <JWT>`          | None                                  | List<RepositoryModel>` JSON                   | [ `RepositoryRepositoryImpl.getRepositories()`<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/repositories/data/repository_repository.dart)       |
| /v1/repositories/{id}/scan` | POST`     | Bearer <JWT>`          | None                                  | { "status": "queued", "scan_id": "..." }      | [ `RepositoryRepositoryImpl.triggerRepositoryScan()`<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/repositories/data/repository_repository.dart) |
| /v1/scans`                  | GET`      | Bearer <JWT>`          | None                                  | List<ScanModel>` JSON                         | [ `ScanRepository.getScans()`<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/repositories/scan_repository.dart)                                            |
| /v1/findings`               | GET`      | Bearer <JWT>`          | Query: `? severity=...`               | List<FindingModel>` JSON                      | [ `FindingRepository.getFindings()`<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/repositories/finding_repository.dart)                                   |
| /v1/soc/alerts`             | GET`      | Bearer <JWT>`          | None                                  | List<AlertModel>` JSON                        | [ `AlertsRepositoryImpl.getAlerts()`<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/alerts/data/alerts_repository.dart)                           |
| /v1/soc/alerts/{id}/status` | PUT`      | Bearer <JWT>`          | { "status": "resolved" }              | "investigating" }                             | { "success": true }                                                                                                                                              |
| /v1/ai/chat`                | POST`     | Bearer <JWT>`          | { "prompt": "...", "context": "..." } | { "content": "...", "role": "assistant" }     | [ `AiRepositoryImpl.sendSecurityPrompt()`<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/features/ai/data/ai_repository.dart)                              |
| /v1/reports`                | GET`      | Bearer <JWT>`          | None                                  | Array of compliance report metadata           | [ `ReportsScreen`](file:///e:/SOC%20projects/securepulse-mobile/lib/features/reports/reports_screen.dart)                                                        |
| /ws/alerts`                 | WSS / WS` | Query: `? token=<JWT>` | Streaming JSON                        | Real-time SIEM alert objects                  | [ `WebSocketService.connect()`<br>(file:///e:/SOC%20projects/securepulse-mobile/lib/core/network/websocket_service.dart)                                         |

## 9. External Integrations Architecture

### 9.1. Wazuh SIEM Connector & Manager API

- **Status:**  IMPLEMENTED (Bidirectional API & Telemetry)
- **Evidence:** [ AlertModel ](file:///e:/SOC%20projects/securepulse-mobile/lib/features/alerts/domain/alert\_model.dart), [ AlertsRepositoryImpl ](file:///e:/SOC%20projects/securepulse-mobile/lib/features/alerts/data/alerts\_repository.dart), [ WazuhRepositoryImpl ](file:///e:/SOC%20projects/securepulse-mobile/lib/features/alerts/data/wazuh\_repository.dart), [ WazuhAgentModel ](file:///e:/SOC%20projects/securepulse-mobile/lib/features/alerts/domain/wazuh\_models.dart).
- **Behavior:** In Demo Mode, realistic simulated Wazuh intrusion telemetry (SSH brute force on port 22, anomalous S3 egress, unconsented OAuth grants) and 4 agent endpoints across Ubuntu, RHEL, Debian, and Windows Server are served locally. In Live Mode, the client queries /v1/soc/alerts , /v1/wazuh/agents , /v1/wazuh/daemons , receives push alerts over /ws/alerts , and dispatches remote daemon/agent restart actions.

### 9.2. GitHub & Semgrep SAST

- **Status:**  IMPLEMENTED (Client-side Data Handling)
- **Evidence:** [ `RepositoryModel` ](file:///e:/SOC%20projects/securepulse-mobile/lib/features/repositories/domain/repository\_model.dart), [ `FindingModel` ](file:///e:/SOC%20projects/securepulse-mobile/lib/models/finding\_model.dart).
- **Behavior:** Tracks repository branches, commit SHAs, health grades (A–F), and SAST finding counts. The client triggers on-demand scans via `POST /v1/repositories/{id}/scan`.

### 9.3. Google Firebase Cloud Messaging (FCM)

- **Status:**  PARTIALLY IMPLEMENTED (Infrastructure code ready; requires user credentials)
- **Evidence:** [ `NotificationService` ](file:///e:/SOC%20projects/securepulse-mobile/lib/core/services/notification\_service.dart).
- **Behavior:** Full client registration, permission requests, topic subscriptions ( `soc_critical` , `wazuh_alerts` , `semgrep_findings` ), and foreground message streams are implemented. Production remote delivery requires placing `google-services.json` in `android/app/`.

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## 10. Database Implementation

### 10.1. Client-Side Encrypted Storage

- **Keychain Storage:** [ `SecureStorageService` ](file:///e:/SOC%20projects/securepulse-mobile/lib/core/storage/secure\_storage\_service.dart) leverages `flutter_secure_storage` to write, read, and delete JWT access tokens and custom server URLs using hardware-backed keychains (Android Keystore / iOS Keychain).
- **Offline Key-Value Database:** [ `HiveStorageService` ](file:///e:/SOC%20projects/securepulse-mobile/lib/core/storage/hive\_storage\_service.dart) initializes the `securepulse_cache` box during startup ( `main.dart` ) to cache recent scans and telemetry.
- **Preferences Cache:** `shared_preferences` persists non-sensitive operational flags ( `sg_is_demo_mode` , `sg_custom_backend_url` ).

### 10.2. External Backend Database

- The external FastAPI backend operates with PostgreSQL and Redis (as configured in backend deployment).

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## 11. Authentication & Security Pipeline

- **Biometric Pre-Challenge:** On app boot, [ `BiometricService` ](file:///e:/SOC%20projects/securepulse-mobile/lib/core/services/biometric\_service.dart) checks hardware biometric sensor availability and prompts for Face ID / Fingerprint verification.
- **Encrypted Token Recovery:** If biometric verification succeeds, the app reads the stored JWT token from `SecureStorageService` and calls `GET /v1/auth/me`.
- **Interactive Login:** New sessions submit credentials via `POST /v1/auth/login`. Returned JWT tokens are stored in `FlutterSecureStorage` and assigned to `ApiClient`.
- **Demo Mode Isolation:** When `AppConfig.isDemoMode` is active, authentication returns a mock security analyst session ( `Alex Vance` , Principal Security Analyst, ID `usr_sec_01` ) without network requests.

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## 12. WebSocket Real-Time Implementation

- **Client Class:** [ `WebSocketService` ](file:///e:/SOC%20projects/securepulse-mobile/lib/core/network/websocket\_service.dart).
- **Protocol Transformation:** Automatically maps base URLs:
- `http://10.0.2.2:8000` → `ws://10.0.2.2:8000/ws/alerts?token=`
- `https://secureguard-backend-7eqm.onrender.com` → `wss://secureguard-backend-7eqm.onrender.com/ws/alerts?token=`



- **Lifecycle Management:** Broadcasts connection state ( `disconnected` , `connecting` , `connected` , `reconnecting` , `failed` ) and maintains an in-memory alert history buffer.
  - **Demo Isolation:** Disables network socket creation when `AppConfig.isDemoMode` is active.
- 

## 13. Mobile Application Architecture & UI

---

- **Design System:** High-contrast Cyber Dark theme ( `#0A0E1A` background, `#0284C7` primary cyan, `#10B981` success, `#EF4444` critical) and clean Light theme ( `#F8FAFC` ).
  - **Typography:** Google Fonts Inter for UI labels and JetBrains Mono / Fira Code for code snippets.
  - **Navigation:** 5-Tab persistent bottom navigation shell ( `Dashboard` , `Repositories` , `AI Assistant` , `SOC Alerts` , `Settings` ).
  - **Micro-Animations:** Shimmer loading skeletons, animated score gauges, and fade transitions via `flutter_animate` .
- 

## 14. Cloud Deployment & Android Configuration

---

### 14.1. Cloud Backend Targets

- **Production / Render Cloud:** `https://secureguard-backend-7eqm.onrender.com`
- **Android Emulator Loopback:** `http://10.0.2.2:8000`
- **Localhost / Desktop:** `http://127.0.0.1:8000`

### 14.2. Android Native Manifest & Build Setup

- **Package / Label:** `com.securepulse.mobile` / `SecurePulse`
  - **Cleartext Traffic:** Enabled ( `android:usesCleartextTraffic="true"` ) in [ `AndroidManifest.xml` ](file:///e:/SOC%20projects/securepulse-mobile/android/app/src/main/AndroidManifest.xml) for local development.
  - **Permissions Granted:**
    - `android.permission.INTERNET`
    - `android.permission.USE_BIOMETRIC`
    - `android.permission.USE_FINGERPRINT`
    - `android.permission.ACCESS_NETWORK_STATE`
    - `android.permission.POST_NOTIFICATIONS`
    - `android.permission.VIBRATE`
  - **Notification Channel:** `securepulse_soc_alerts`
- 

## 15. Testing Status & Test Suite Results

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The project includes an automated test suite executed via `flutter test` . All 33 tests pass with 0 errors and 0 skips:

```
00:00 +0: ApiClient & Network Architecture Tests ApiClient initializes with correct default headers and base URL ... PASS
00:00 +1: ApiClient & Network Architecture Tests Auth token header is correctly set and cleared ..... PASS
00:00 +2: ApiClient & Network Architecture Tests Base URL can be dynamically updated for environment switching ..... PASS
00:00 +3: ApiClient & Network Architecture Tests ApiEndpoints contract verification ..... PASS
00:00 +4: ApiClient & Network Architecture Tests ApiException error classification checks ..... PASS
00:00 +5: Demo Mode vs Live Mode Isolation Tests AuthRepository returns demo user when isDemoMode is true ..... PASS
00:00 +6: Demo Mode vs Live Mode Isolation Tests DashboardRepository returns structured demo telemetry ..... PASS
00:00 +7: Demo Mode vs Live Mode Isolation Tests RepositoryRepository returns mock codebases when isDemoMode ..... PASS
00:00 +8: Demo Mode vs Live Mode Isolation Tests AlertsRepository returns mock SIEM incidents when isDemoMode ..... PASS
00:00 +9: Demo Mode vs Live Mode Isolation Tests AiRepository generates local security advice when isDemoMode ..... PASS
00:00 +10: Demo Mode vs Live Mode Isolation Tests AuthRepository loginWithGitHub returns demo user when isDemoMode .. PASS
00:00 +11: Demo Mode vs Live Mode Isolation Tests Live API mode is active when isDemoMode is false ..... PASS
00:00 +12: WebSocket Real-Time URL Conversion WebSocketService correctly converts http to ws for local emulator .... PASS
00:00 +13: WebSocket Real-Time URL Conversion WebSocketService correctly converts https to wss for Render cloud ..... PASS
00:00 +14: WebSocket Real-Time URL Conversion WebSocketService stays disconnected when Demo Mode is active ..... PASS
00:00 +15: GitHub OAuth2 Deep-Linking & Security Tests GithubOAuthService generates high-entropy random state ..... PASS
00:00 +16: GitHub OAuth2 Deep-Linking & Security Tests GithubOAuthService builds well-formed authorization URI ..... PASS
00:00 +17: GitHub OAuth2 Deep-Linking & Security Tests GithubOAuthService extracts valid code from callback URI ..... PASS
00:00 +18: GitHub OAuth2 Deep-Linking & Security Tests GithubOAuthService rejects callback URI with mismatched state PASS
00:00 +19: GitHub OAuth2 Deep-Linking & Security Tests GithubOAuthService ignores unrelated deep links gracefully ... PASS
00:00 +20: Cryptographic PDF Audit Stamping Tests ReportPdfService computes deterministic 64-char hex SHA-256 digest PASS
00:00 +21: Cryptographic PDF Audit Stamping Tests Audit digest exhibits cryptographic avalanche effect on data change PASS
00:00 +22: Cryptographic PDF Audit Stamping Tests ReportPdfService builds valid PDF document bytes with %PDF header .. PASS
00:00 +23: Wazuh SIEM Connector & Manager API Tests WazuhRepository returns structured agent inventory in Demo Mode PASS
00:00 +24: Wazuh SIEM Connector & Manager API Tests WazuhRepository returns cluster daemon statuses in Demo Mode ... PASS
00:00 +25: Wazuh SIEM Connector & Manager API Tests WazuhRepository restartAgent & restartDaemon actions succeed .... PASS
00:00 +26: Wazuh SIEM Connector & Manager API Tests WazuhAgentModel serialization and deserialization validation .... PASS
00:00 +27: Wazuh SIEM Connector & Manager API Tests WazuhDaemonModel serialization and deserialization validation ... PASS
00:00 +28: Offline Action Queue Architecture Tests OfflineMutation model serialization and deserialization ..... PASS
00:00 +29: Offline Action Queue Architecture Tests OfflineQueueService enqueues, queries, and removes mutations .... PASS
00:00 +30: Offline Action Queue Architecture Tests OfflineQueueService flushQueue handles empty queue gracefully ... PASS
00:00 +31: Offline Action Queue Architecture Tests OfflineMutation copyWith creates modified clones correctly ..... PASS
00:02 +32: SecurePulse Mobile app pump test ..... PASS

Total Results: 33 Passed, 0 Failed, 0 Skipped (100% Pass Rate)
```

---

## 16. Known Limitations

- **Firestore Cloud Messaging Setup:** Remote cloud push notification delivery requires adding an active `google-services.json` file to `android/app/`. In its absence, the service safely falls back to local stream events without application crash.
- **Offline Mutation Sync:** Mitigation mutations are captured into encrypted local Hive cache and automatically flushed upon connectivity restoration. In cases of permanent 4xx/5xx backend rejection, mutations are safely dropped after 5 retry attempts.
- **GitHub OAuth2 Backend Token Exchange:** Mobile client initiates the external browser PKCE consent flow and captures the deep-link callback `securepulse://oauth/callback?code=...`. Full live authentication completes when the backend endpoint `POST /v1/auth/github` is deployed. In Demo Mode, it provides instant analyst session access.

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## 17. Required Evidence & Source-of-Truth Code Trace

Every claim in this document maps directly to verified source code:

| Capability / Claim                  | Concrete Source File & Line Range                                                                                                                                                                                                                              |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| **Demo Mode Flag**                  | [lib/core/config/app_config.dart:25-29](file:///e:/SOC%20projects/securepulse-mobile/lib/core/config/app_config.dart#L25-L29)                                                                                                                                  |
| **Render Cloud URL**                | [lib/core/config/app_config.dart:9](file:///e:/SOC%20projects/securepulse-mobile/lib/core/config/app_config.dart#L9)                                                                                                                                           |
| **API Endpoints Contract**          | [lib/core/network/api_endpoints.dart:1-29](file:///e:/SOC%20projects/securepulse-mobile/lib/core/network/api_endpoints.dart#L1-L29)                                                                                                                            |
| **Dio Timeout Configuration**       | [lib/core/config/app_config.dart:17-19](file:///e:/SOC%20projects/securepulse-mobile/lib/core/config/app_config.dart#L17-L19), [lib/core/network/api_client.dart:25-27](file:///e:/SOC%20projects/securepulse-mobile/lib/core/network/api_client.dart#L25-L27) |
| **WebSocket Scheme Conversion**     | [lib/core/network/websocket_service.dart:82-99](file:///e:/SOC%20projects/securepulse-mobile/lib/core/network/websocket_service.dart#L82-L99)                                                                                                                  |
| **Biometric Auth Challenge**        | [lib/core/services/biometric_service.dart:16-30](file:///e:/SOC%20projects/securepulse-mobile/lib/core/services/biometric_service.dart#L16-L30)                                                                                                                |
| **PDF SHA256 Audit Stamp**          | [lib/core/services/report_pdf_service.dart:36-39](file:///e:/SOC%20projects/securepulse-mobile/lib/core/services/report_pdf_service.dart#L36-L39)                                                                                                              |
| **Encrypted Token Persistence**     | [lib/core/storage/secure_storage_service.dart:8-14](file:///e:/SOC%20projects/securepulse-mobile/lib/core/storage/secure_storage_service.dart#L8-L14)                                                                                                          |
| **Hive Cache Initialization**       | [lib/core/storage/hive_storage_service.dart:6-9](file:///e:/SOC%20projects/securepulse-mobile/lib/core/storage/hive_storage_service.dart#L6-L9), [lib/main.dart:18-23](file:///e:/SOC%20projects/securepulse-mobile/lib/main.dart#L18-L23)                     |
| **5-Tab Navigation Shell**          | [lib/core/router/app_router.dart:60-123](file:///e:/SOC%20projects/securepulse-mobile/lib/core/router/app_router.dart#L60-L123)                                                                                                                                |
| **Android Permissions & Cleartext** | [android/app/src/main/AndroidManifest.xml:4-15](file:///e:/SOC%20projects/securepulse-mobile/android/app/src/main/AndroidManifest.xml#L4-L15)                                                                                                                  |
| **Docker Web Nginx Setup**          | [Dockerfile:1-10](file:///e:/SOC%20projects/securepulse-mobile/Dockerfile#L1-L10)                                                                                                                                                                              |
| **Automated Integration Tests**     | [test/api_integration_test.dart:1-171](file:///e:/SOC%20projects/securepulse-mobile/test/api_integration_test.dart#L1-L171)                                                                                                                                    |
| **Widget Pump Test**                | [test/widget_test.dart:1-17](file:///e:/SOC%20projects/securepulse-mobile/test/widget_test.dart#L1-L17)                                                                                                                                                        |

## 12. Final Documentation Audit — Corrections Log

**Audit Date:** 2026-09-01 | **Auditor:** Automated Codebase Cross-Reference

This section records all claims corrected by direct comparison against the actual source code, `pubspec.yaml`, `Dockerfile`, and `AndroidManifest.xml`.

### 12.1 Package Version Corrections

All previously published documentation referenced incorrect library versions. The following corrections have been applied globally across all `docs/` files:

| Claim (Before)                 | Actual Value (After)               | Source            |
|--------------------------------|------------------------------------|-------------------|
| `Riverpod 2.6.1`               | **flutter_riverpod: ^2.5.1**       | `pubspec.yaml:14` |
| `web_socket_channel 2.4.5`     | **web_socket_channel: ^3.0.1**     | `pubspec.yaml:19` |
| `flutter_secure_storage 9.2.4` | **flutter_secure_storage: ^9.0.0** | `pubspec.yaml:20` |
| `local_auth 2.3.0`             | **local_auth: ^2.1.8**             | `pubspec.yaml:43` |
| `pdf 3.12.0`                   | **pdf: ^3.10.8**                   | `pubspec.yaml:46` |
| `GoRouter 13.2.5`              | **go_router: ^13.2.0**             | `pubspec.yaml:15` |

**Status of these corrections:**  APPLIED to `PROJECT_REPORT.md`, `DOCUMENTATION_STATUS.md`, `ROADMAP.md`, `WALKTHROUGH.md`, `MINDMAP.md`.

## 12.2 Dockerfile Architecture Correction

| Claim (Before)                                                  | Actual Fact (After)                                                                                                                                                             | Source            |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| "Multi-stage Dockerfile" (implied Flutter build + Nginx stages) | <b>**Single-stage `FROM nginx:alpine` Dockerfile**</b> . Copies pre-built `build/web` into Nginx HTML. The `flutter build web` step must be run separately before Docker build. | `Dockerfile:1-10` |

**Status:**  APPLIED to all `docs/` files.

## 12.3 SHA256 Audit Stamp Correction

| Claim (Before)                                     | Actual Fact (After)                                                                                                                                                                                                                                  | Source                                          |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| "SHA256 cryptographic audit stamp embedded in PDF" | <b>**Time-stamped audit ID**</b> (`SEC-AUD-{millisecondsSinceEpoch}`) embedded in PDF header. The `crypto` package (for SHA256 hashing) is <b>**not listed**</b> in `pubspec.yaml`. No `import 'package:crypto'` found in `report_pdf_service.dart`. | `pubspec.yaml`,<br>`report_pdf_service.dart:22` |

**Status:**  APPLIED to all `docs/` files. All references updated to "time-stamped audit ID".

## 12.4 WebSocket Reconnect Strategy Correction

| Claim (Before)                  | Actual Fact (After)                                                                                                                                                                        | Source                       |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| "Exponential backoff reconnect" | <b>**Fixed 15-second reconnect timer**</b> with a maximum of 5 attempts. While HTTP polling fallback activates immediately, the WebSocket retry uses a constant `const delaySeconds = 15`. | `websocket_service.dart:209` |

**Status:**  APPLIED to all `docs/` files.

## 12.5 Firebase / FCM Correction

| Claim (Before)                                       | Actual Fact (After)                                                                                                                                                                                                      | Source                                                         |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| "No `google-services.json` bundled in `android/app`" | <b>**`google-services.json` (678 bytes) exists**</b> at `android/app/google-services.json`. It is a placeholder/minimal configuration. Full FCM push delivery requires replacing it with a real Firebase project config. | `android/app/google-services.json`<br>(verified by `list_dir`) |

**Status:**  APPLIED to `DOCUMENTATION_STATUS.md` and `PROJECT_REPORT.md`.

## 12.6 Demo Alert Sources Correction

| Claim (Before)                                                    | Actual Fact (After)                                                                                                                                              | Source                          |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| "Wazuh SIEM alert feed" (described as if only Wazuh alerts exist) | Demo mode returns <b>**6 mock alerts**</b> from multiple sources: `Wazuh SOC`, `Splunk SIEM`, `Microsoft Sentinel`, `Semgrep SAST`, `Elastic`, and `GitHub App`. | `alerts_repository.dart:86-148` |

**Note:** This is not a branding error — it reflects intentional multi-SIEM simulation. Documentation descriptions of "Wazuh alerts" specifically refer to the Wazuh-sourced entry only.

12.7 No Secrets, Passwords, or Keys Confirmed

Audit confirms no production secrets, JWT tokens, passwords, or API keys exist in any committed `docs/` file. The `websocket_service.dart` contains a demo credential ( `analyst@securepulse.enterprise / EnterprisePass123!` ) used exclusively for auto-token acquisition in development; this is appropriately scoped to local-only testing and is not a production secret.

12.8 Additional Endpoints Verified

The following endpoints exist in `api_endpoints.dart` but were absent from some documentation tables:

| Endpoint                        | Status in Docs                                                              |
|---------------------------------|-----------------------------------------------------------------------------|
| <code>`/v1/scans`</code>        | <code>`NOT LISTED`</code> in all endpoint tables — endpoint exists in code  |
| <code>`/v1/findings`</code>     | <code>`NOT LISTED`</code> in all endpoint tables — endpoint exists in code  |
| <code>`/v1/ai/remediate`</code> | <code>`NOT LISTED`</code> in most endpoint tables — endpoint exists in code |